GEOMETRIC GRAPH RESEARCH / BOUNDARY INVESTIGATION

The boundary is different

Exact outer triangles, a derived area constant, and the rare cells that defeat global moment scaling

16 September 2026

The central change. The local Poisson-line description developed for fixed interior locations does not describe the outermost cells. Every convex-hull edge of a generic complete straight-line graph borders an exact triangular face. Under smooth-domain uniform sampling, these boundary triangles have an area scale of order n^{-1}, rather than the n^{-4} bulk scale.

A parameter-free boundary prediction. A parabolic empty-cone calculation yields a proposed asymptotic ratio

$$\frac{\mathbb{E}[\text{total area of hull-edge-adjacent faces}]}{\mathbb{E}[\text{area missing outside the random hull}]} \longrightarrow 0.1162721743\ldots.$$

The constant was evaluated from its defining integral before the production simulations, not calibrated to them.

Independent checks. This phase constructs the exact boundary faces of 12,000 independent point configurations, up to 32,768 cities, obtaining 438,947 boundary triangles. A second experiment checks an exact center-classification formula in 2.4 million independent angular trials. Full-graph polygonization validates 772 triangles from 130 small graphs; another 480 geometric checks validate the center classification.

Status. The deterministic propositions and center probabilities below have direct proofs. The boundary expectation is a derived asymptotic result with a localization argument, rather than an independently reviewed theorem or an established novelty claim. Classical random-hull theory supplies the parabolic framework and missing-area asymptotics. Numerical agreement is evidence, not a substitute for independent proof review.

Reproducibility. The companion archive includes all new code, graph-level data, deterministic seeds, sampled face records, quadrature checks, seven figures, and this LaTeX source. Earlier research archives are not duplicated.

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1 Scope: which boundary and which cells?

Let $P = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$, with $n \geq 4$, be in general position. Draw every segment $[p_i, p_j]$. The bounded components of the resulting subdivision are its faces. Let $K(P) = \mathrm{conv}(P)$ have area H and h edges.

A *boundary face* in this note means a bounded face sharing a positive-length edge with $\partial K(P)$. A cell that merely touches a hull vertex is not counted. Nor is every visually peripheral cell a boundary face in this sense.

For a sampling domain D with area S, define

$$B_n = \sum_{f \text{ boundary}} A_f, \qquad M_n = S - H_n.$$

Here B_n is area *inside* the random hull, occupied by its edge-adjacent faces; M_n is area *outside* the hull but inside the sampling domain. The principal ratio is $\mathbb{E}B_n/\mathbb{E}M_n$, not $\mathbb{E}(B_n/M_n)$.

The earlier interior limit fixes a location independently of the point sample, strictly away from the boundary, and magnifies lengths by order n^2. It does not give uniform control as that location approaches the random hull. The present investigation does not assert a full boundary limit for all chords: many chords retain shared-endpoint dependence there. It instead isolates exact boundary faces and the finite empty-region events determining them.

2 Every hull edge borders a triangular face

Proposition 1 (Deterministic boundary triangle)**.** *For a point set in general position with at least four points, each hull edge borders exactly one bounded triangular face. Different hull edges have different such faces. Consequently there are exactly h boundary faces.*

Proof. Use an orientation-preserving rigid motion to place a chosen hull edge at $u = (0, 0)$, $v = (L, 0)$, with all other cities (x_i, y_i) satisfying $y_i > 0$. Define

$$a = \max_i \frac{x_i}{y_i}, \qquad b = \max_i \frac{L - x_i}{y_i}.$$

For every remaining city, $x_i \leq a y_i$ and $x_i \geq L - b y_i$. The two supporting rays meet at

$$q = \left(\frac{aL}{a+b}, \frac{L}{a+b} \right).$$

The denominator is positive: for each city, the sum of the corresponding two ratios is $L/y_i > 0$.

The open triangle uvq is described by $y > 0$, $x > ay$, and $x < L - by$. Every other city is on the opposite side of both supporting lines. Segments joining two such cities stay in their common convex wedge; segments incident to u or v cannot enter the open triangle. The two supporting segments reach q, since the supporting cities have $y_i \geq L/(a+b)$. Thus uvq is exactly the adjacent face.

A single triangular face cannot contain two hull edges when $n \geq 4$ and no three cities are collinear. If it did, its three vertices would be the consecutive hull vertices supporting those edges. Any fourth city would force at least one of the two angular extremizers inward, contradicting that triangle's emptiness. This also follows from the two-essential-city characterization proved below. □

The calculation is constructive:

$$\boxed{A_{uv} = \frac{L^2}{2(a+b)}.} \tag{1}$$

The inradius is obtained from $r = 2A/P$. After the convex hull has been found, all boundary faces require $O(nh)$ arithmetic operations. The complete graph and its order-n^4 number of crossings need not be constructed.

This efficiency is why the experiments can reach 32,768 cities. They construct the actual outer triangles, *not* every interior face. Huge all-face counts later in the note are expectation formulas, not claims that those trillions of polygons were explicitly built.

2.1 The original seven-city picture

For the original coordinates $A = (6, 4)$, $B = (1, 4)$, $C = (1, 0)$, $D = (4, 8)$, $E = (6, 3)$, $F = (0, 7)$, $G = (2, 11)$, the six hull edges yield:

Hull edge	Adjacent triangular area
FC	2
CE	2
EA	0.15151515…
AD	1/3
DG	0.4375
GF	2.59259259…

The total is 7.51494108… out of hull area 38, or 19.77616%. The giant lower-central cell from the early regressions is *not* one of these hull-edge-adjacent faces; the distinction is intentional.

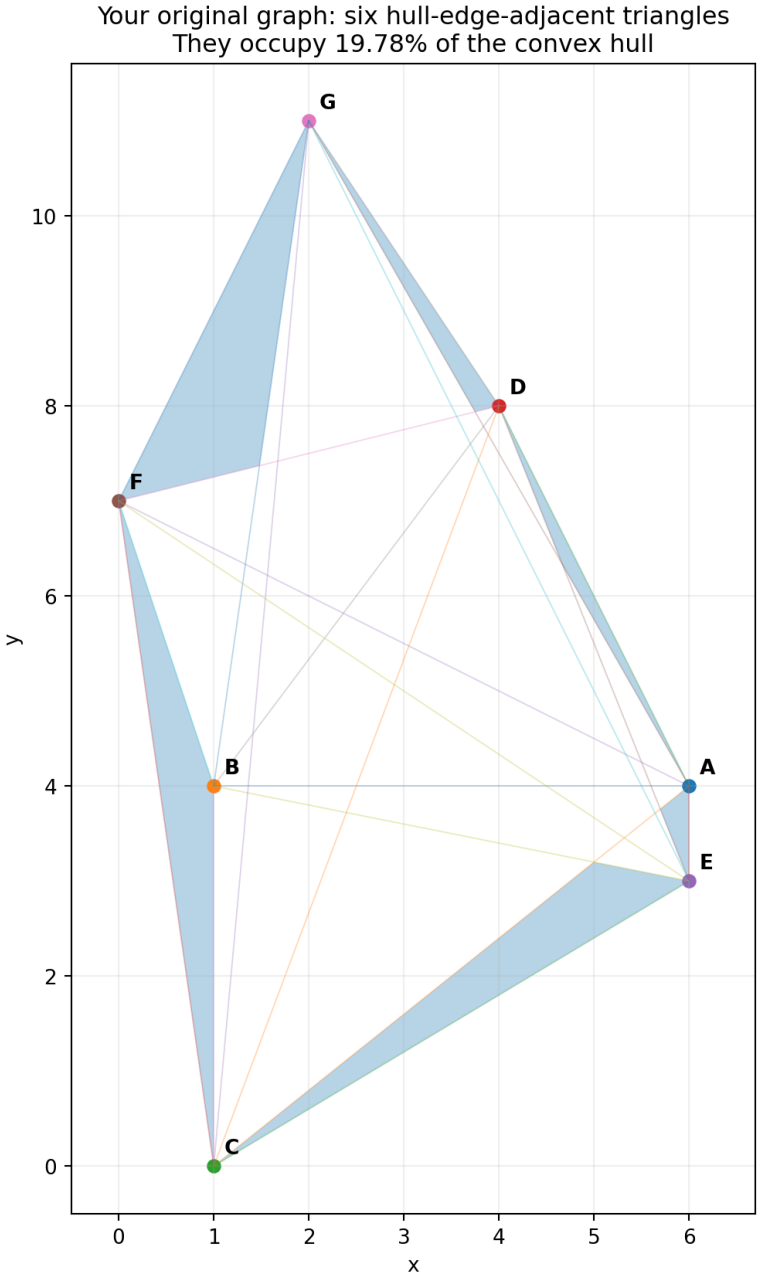


Figure 1: The exact boundary class in the original graph. Interior faces, including large ones, remain unfilled.

2.2 Independent geometric validation

For 130 random graphs with 4 through 16 cities, the supporting-ray construction was compared with full segment noding and polygonization. All 772 matched hull-edge-adjacent faces were triangles. The worst relative symmetric-difference area was 5.29×10^{-13}. The test does not replace the proof, but independently checks the implementation and the finite-segment interpretation.

3 An exact center law related to Wendel's theorem

Fix a query point x, not on a graph edge. When $x \in K(P)$, call a city *essential for* x if deleting it removes x from the convex hull. Let $J(x)$ count these cities.

Lemma 1 (Two essential cities identify the outer triangle)**.** *For $n \geq 4$, a query strictly inside the hull has $J(x) \leq 2$. It lies in a boundary face if and only if $J(x) = 2$.*

Proof. Carathéodory's theorem gives a triangle of cities containing x. All essential cities belong to it. If three were essential, insert any fourth city into this representation: a barycentric elimination replaces at least one original vertex by the fourth city while retaining containment. This contradicts three-fold essentiality.

Suppose u, v are essential, and uvw contains x. For every other city z, applying the same four-point elimination shows uvz must contain x: a representation omitting u or v is forbidden. Thus every other city lies in

$$Q_x(u, v) = \{z : x \in \text{conv}(u, v, z)\},$$

an unbounded cone with apex x. This is exactly the angular condition for x to lie in the boundary triangle of hull edge uv. Conversely, in such a triangle both hull endpoints are necessary for containing x after a one-city deletion. □

Proposition 2 (Center classification)**.** *Let $n \geq 4$ cities be iid from a centrally symmetric distribution about 0, with a continuous nondegenerate angular distribution. Then*

$$\mathbb{P}(0 \notin K_n) = \frac{n}{2^{n-1}}, \tag{2}$$

$$\mathbb{P}(0 \textit{ lies in a boundary face}) = \frac{n}{2^{n-1}}, \tag{3}$$

$$\mathbb{P}(0 \in K_n,\ J(0) = 1) = \frac{n(n-4)}{2^{n-1}}, \tag{4}$$

$$\mathbb{P}(0 \in K_n,\ J(0) = 0) = 1 - \frac{n(n-2)}{2^{n-1}}. \tag{5}$$

The first identity is the planar case of Wendel's classical theorem [1]. The boundary-face companion is derived here; no claim that it is absent from prior literature is made.

Proof. Write the angular cumulative distribution as F. Central symmetry gives $F(\theta + \pi) = F(\theta) + 1/2$. Transforming the angles by F produces iid uniform angular mass coordinates and preserves semicircle containment.

For a chosen unordered pair, the cone $Q_0(u, v)$ has angular probability mass Q uniformly distributed on $[0, 1/2]$. The other $n - 2$ cities all lie in it with probability Q^{n-2}. Since a boundary face containing the query has a unique hull pair,

$$\mathbb{P}(J = 2) = \binom{n}{2}\mathbb{E}Q^{n-2} = \binom{n}{2}\frac{1}{(n-1)2^{n-2}} = \frac{n}{2^{n-1}}.$$

This also explains why radial distances play no role.

By exchangeability,

$$\mathbb{E}[J\,1_{\{0\in K_n\}}] = n\,[\mathbb{P}(0 \in K_n) - \mathbb{P}(0 \in K_{n-1})] = \frac{n(n-2)}{2^{n-1}}.$$

Subtracting twice the two-essential probability gives the one-essential probability, and subtraction from the total gives the last formula. □

For example, at $n = 8$ the center is outside with probability 1/16, inside a boundary triangle with probability 1/16, inside with one essential city with probability 1/4, and robust to every single-city deletion with probability 5/8.

3.1 Checks across nonuniform angular distributions

The center law was checked in 100,000 independent trials for each of eight city counts and three symmetric distributions: isotropic, a 6:1 elliptical Gaussian, and two symmetric Gaussian clusters. This gives 2.4 million angular trials. All inside queries had $J \le 2$.

For a separate 480 point configurations, the classification from circular angular gaps was compared directly with membership in the geometrically constructed boundary triangles. There were no mismatches. The angular trials do not build millions of full subdivisions, and are not described as such.

4 The correct boundary magnification is anisotropic

Assume now that the cities are iid uniform in a compact convex domain D with a C^3 boundary and strictly positive curvature. Let $S = \text{area}(D)$ and let $\kappa(s)$ be curvature at boundary arc-length coordinate s.

At a fixed boundary point, use tangent coordinate u and inward-normal depth z. Locally,

$$z \ge \frac{\kappa u^2}{2} + O(|u|^3).$$

Choose scales

$$a_n = \left(\frac{S}{n\kappa}\right)^{1/3}, \qquad b_n = \kappa^{1/3}\left(\frac{S}{n}\right)^{2/3}. \tag{6}$$

Then $a_n b_n = S/n$ and $\kappa a_n^2 = b_n$. The transformed city cloud in coordinates $(U, Z) = (u/a_n, z/b_n)$ converges on bounded sets to a unit-intensity Poisson point process on

$$\mathcal{P} = \{(U, Z) : Z \ge U^2/2\}.$$

The parabolic random-hull framework is classical; see Calka–Schreiber–Yukich [2]. Their results should not be confused with a claim that every complete-graph chord becomes independent near the boundary.

The scale calculation immediately distinguishes the regimes:

Regime	Linear scales	Area scale
Fixed interior bulk	n^{-2} in both directions	n^{-4}
Smooth hull boundary	$n^{-1/3}$ tangent; $n^{-2/3}$ normal	n^{-1}

The boundary statement is an ensemble scale for the outermost triangles, not a claim that every triangle has the same shape or that the largest face is necessarily a hull-edge-adjacent face.

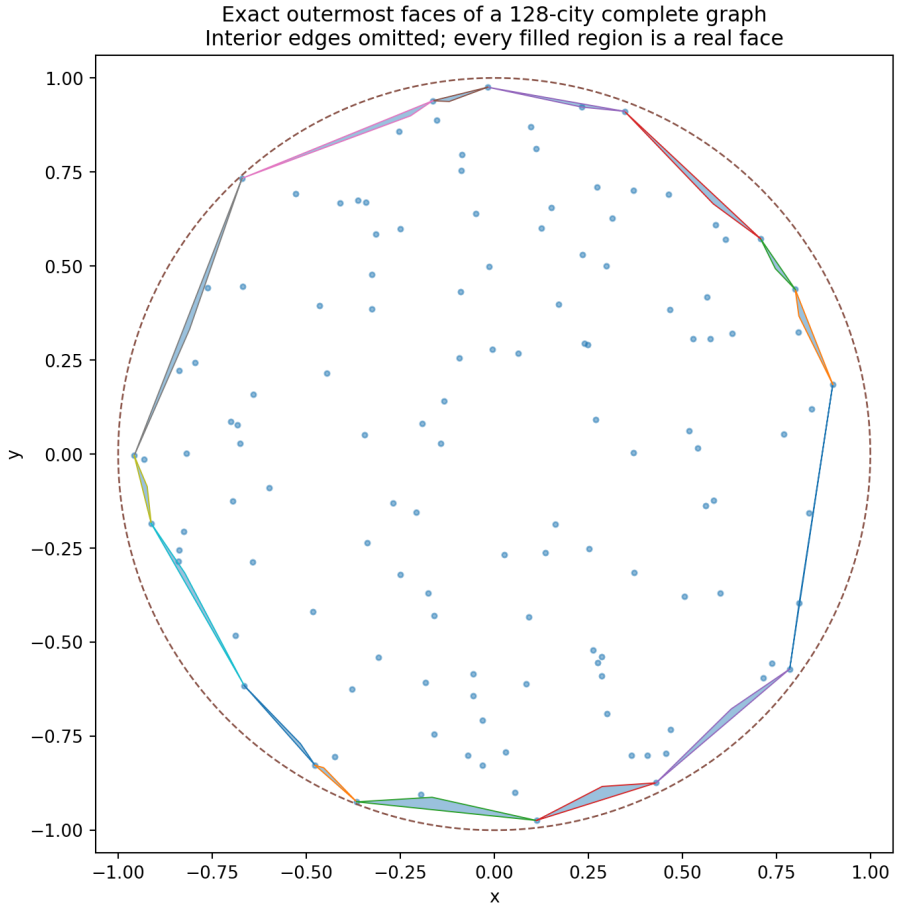


Figure 2: Exact hull-edge-adjacent faces of one 128-city graph. The disk boundary is dashed; the unfilled exterior gaps contribute to M_n. The interior complete-graph edges are omitted only for clarity.

5 A closed-form limiting distribution for the radial hull gap

Here the parabolic model yields a full distribution, not only a mean.

Let a homogeneous Poisson process of intensity λ live on $\mathcal{P}$. Its lower hull has one edge crossing the vertical axis. Write that edge's line as

$$Z = c + sU, \qquad c > 0.$$

The area of the parabolic cap below the line is

$$V(c, s) = \frac{2}{3}(s^2 + 2c)^{3/2}.$$

Write its two endpoint abscissas as $u < 0 < v$. The endpoint-to-line change of variables has Jacobian $v - u$. Integrating over allowed endpoints gives

$$\int_{s-\sqrt{s^2+2c}}^{0} \int_{0}^{s+\sqrt{s^2+2c}} (v - u)\, \mathrm{d}v\, \mathrm{d}u = 2c\sqrt{s^2 + 2c}.$$

The Poisson two-point formula and cap-emptiness probability give the joint density

$$\boxed{f(c, s) = 2\lambda^2 c\sqrt{s^2 + 2c}\, \exp\left[-\frac{2\lambda}{3}(s^2 + 2c)^{3/2}\right].} \tag{7}$$

Its integral is one, corresponding to the unique lower hull edge crossing the axis.

Set $R = \sqrt{s^2 + 2c}$ and write $s = R\cos\theta$, $\sqrt{2c} = R\sin\theta$. The density factors into a radial part and an angular part. Equivalently,

$$\boxed{C \overset{d}{=} \frac{1}{2}\left(\frac{3Z}{2\lambda}\right)^{2/3} B, \quad Z \sim \text{Gamma}(2,1), \quad B \sim \text{Beta}(2,1/2),} \tag{8}$$

with Z and B independent. This is a representation of the chosen parabolic radial-gap law; related local radius-function distributions belong to the classical random-hull theory [2].

For n uniform cities in a unit-radius disk, let Δ_n be the inward gap between $(0,1)$ and the hull along the vertical axis. With the simpler scaling $(U, Z) = (n^{1/3}x, n^{2/3}(1-y))$, the limiting intensity is $\lambda = 1/\pi$. Hence

$$n^{2/3}\Delta_n \Rightarrow \frac{1}{2}\left(\frac{3\pi Z}{2}\right)^{2/3} B.$$

The limiting mean is $1.6916144829\ldots$. At finite n an axis can miss the hull; the implementation records this rather than resampling. There were no such failures in the production disk experiment.

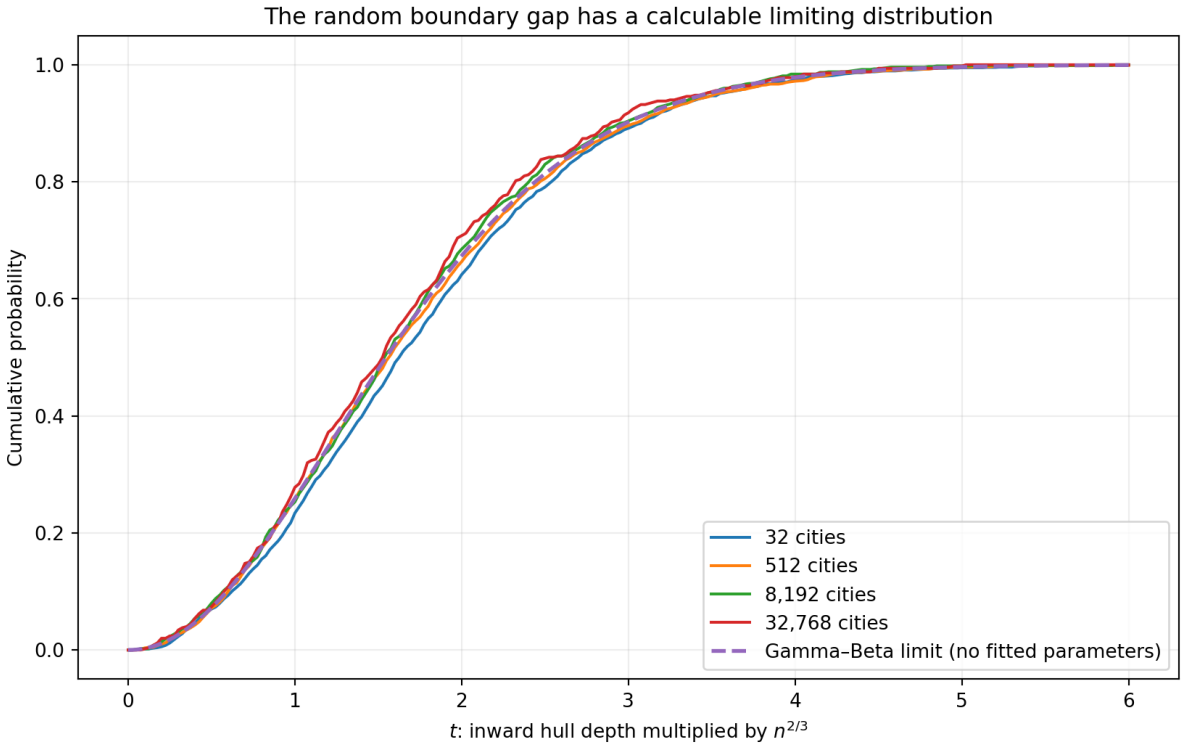


Figure 3: CDFs of the observed disk gap versus the derived Gamma–Beta reference. No shape or rate parameter is fitted. The largest-n group has only 500 independent graphs, explaining visible empirical variation.

At $n = 32, 128, 512, 2048, 8192, 32768$, the empirical maximum CDF discrepancies are respectively $0.0404, 0.0238, 0.0157, 0.0195, 0.0200, 0.0410$. These values are not monotone, because independent group sizes differ. The curves are a distributional check, not proof of a finite-n identity.

6 Deriving the outer-triangle area constant

6.1 The exact finite-sample empty-cone formula

For any interior query x, the boundary characterization gives an exact expression:

$$\mathbb{P}(x \text{ in a boundary face}) = \binom{n}{2} \int_D \int_D \left(\frac{\text{area}(Q_x(u,v) \cap D)}{S} \right)^{n-2} \frac{\mathrm{d}u\, \mathrm{d}v}{S^2}. \tag{9}$$

The pair event means all other cities lie in the cone from which the triangle uvz contains x. Degenerate pairs and edge-boundary queries have measure zero. Integrating (9) over x gives $\mathbb{E}B_n$.

This formula avoids a regression and also avoids transferring an unbounded moment merely from weak point-process convergence.

6.2 Boundary membership in the unit-intensity parabolic model

Take the query $(0,t)$ with $t > 0$. A candidate hull pair has its left endpoint on a line of slope a and its right endpoint on a line of slope b, with $a > b$. The cone containing all remaining cities is

$$Z \geq \max(t + aU, t + bU).$$

Define

$$R_+(s,t) = s + \sqrt{s^2 + 2t}, \qquad R_-(s,t) = \sqrt{s^2 + 2t} - s,$$

and the right-hand parabolic cap area

$$J_+(s,t) = \int_0^{R_+(s,t)} (t + sU - U^2/2)\, \mathrm{d}U = \frac{R_+(s,t)^2 \big(2\sqrt{s^2 + 2t} - s\big)}{6}.$$

The left cap area is $J_+(-s,t)$. Endpoint integration along the two rays contributes $R_-(a,t)^2/2$ and $R_+(b,t)^2/2$. Thus the limiting membership probability is

$$p_\partial(t) = \frac{1}{4} \int_{a>b} R_-(a,t)^2 R_+(b,t)^2 e^{-J_+(a,t) - J_+(-b,t)}\, \mathrm{d}a\, \mathrm{d}b. \tag{10}$$

This is an exact formula within the parabolic Poisson model. The two-point formula counts the unique qualifying hull edge. It does not model the graph crossings as independent Poisson points.

A boundary face with both endpoints on the same tangent side of the query requires a cone excluding the inward normal direction. In the finite smooth domain, that excludes a fixed positive-area region of the bulk, and the contribution is exponentially small. This is why the straddling pairs above give the leading boundary asymptotic.

6.3 An absolutely convergent two-dimensional integral

Set $a = \sqrt{2t} \sinh \alpha$, $b = \sqrt{2t} \sinh \beta$, and then

$$m = (\alpha + \beta)/2, \qquad d = (\alpha - \beta)/2 > 0.$$

Let

$$S(m,d) = \frac{e^{3d} \cosh(3m) + 3e^d \cosh(m)}{6}.$$

The integral over t can now be evaluated using the Gamma function. It yields

$$c_\partial := \int_0^\infty p_\partial(t)\,\mathrm{d}t = \frac{\Gamma(8/3)}{12}\,I, \tag{11}$$

$$I = \int_0^\infty \int_{-\infty}^\infty \frac{e^{-4d}[\cosh(2m)+\cosh(2d)]}{S(m,d)^{8/3}}\,\mathrm{d}m\,\mathrm{d}d. \tag{12}$$

The integrand decays exponentially in $|m|$ and d. Deterministic Gauss–Legendre quadrature, with increasing order and extended cutoffs, gives

$$I = 0.7313263204\ldots, \qquad \boxed{c_\partial = 0.09169463796\ldots.}$$

This numerical constant comes from a specified integral. It is not the intercept of a fit.

6.4 Returning to a smooth domain

Write the affine perimeter as

$$\mathrm{as}(D) = \int_{\partial D} \kappa(s)^{1/3}\,\mathrm{d}s.$$

The local normal scale in (6) leads to the proposed boundary expectation asymptotic

$$\boxed{\mathbb{E}B_n \sim c_\partial S^{2/3}\,\mathrm{as}(D)\,n^{-2/3}.} \tag{13}$$

For comparison, the classical smooth random-hull missing-area asymptotic has the same geometric factor:

$$\mathbb{E}M_n \sim c_M S^{2/3}\,\mathrm{as}(D)\,n^{-2/3}, \qquad c_M = (2/3)^{1/3}\Gamma(5/3) = 0.7886206526\ldots. \tag{14}$$

The connection between hull volume and expected vertex count is also given exactly by Efron's identity; see [3, 4]. Random-hull asymptotics and parabolic scaling are classical inputs [2, 5].

Dividing (13) by (14) gives

$$\boxed{\frac{\mathbb{E}B_n}{\mathbb{E}M_n} \longrightarrow \frac{c_\partial}{c_M} = 0.1162721743\ldots.} \tag{15}$$

The area and curvature factors cancel. This is the central parameter-free prediction tested in this note.

6.5 Localization argument and its status

For completeness, the derivation from the binomial sample can be organized as follows. Start from (9), rather than substituting an expected face count for a random one. Split queries into a fixed interior region and a boundary tube. On the interior region, any qualifying cone omits a region of area bounded below, so its probability is exponentially small. At the boundary, cones not containing the inward normal have the same property.

For the remaining cones, use local coordinates (6). On compact scaled sets, the boundary Taylor expansion converges to the parabola and the binomial void factor $(1 - v/n)^{n-2}$ converges to e^{-v}. The ray Jacobians converge to those in (10). To pass to the integrated expectation, tails must also be controlled. Positive curvature bounds small cap areas above and below by constants times depth to the $3/2$ power. The endpoint integrations contribute only polynomial factors; the void probabilities give the corresponding exponential cap-tail suppression. Regions outside a fixed local coordinate chart have a positive omitted area and contribute exponentially little. These bounds provide the route to dominated convergence in the mean formula, and the limiting integral (11) is absolutely convergent.

Qualification. This is a research-note localization argument, not an independently checked publication-level proof with all uniform constants written out. The constant is exact for the limiting empty-cone calculation. The mean asymptotic for original complete graphs is proposed on the basis of this argument and tested independently below. No full chord-process boundary convergence, variance theorem, or universal rate of convergence is claimed. Polygonal domains with corners and vanishing-curvature boundary points are outside the stated smooth positive-curvature scope.

7 Independent numerical tests

7.1 No parameters trained on the production sample

The quadrature calculation was run before the production experiment. The production data use a fixed independent seed family and no rejection for near-coincident cities. All city samples are iid uniform in their stated domain.

The disk experiment has 9,000 graphs, with $n = 32, 128, 512, 2048, 8192, 32768$. The nonelliptical smooth-domain experiment has 3,000 graphs. Its support function is

$$h(\theta) = 1 + 0.07\cos(3\theta),$$

with radius of curvature $h + h'' = 1 - 0.56\cos(3\theta) > 0$. Its exact area is $S = \pi(1 - 4(0.07)^2)$ and its affine perimeter is $6.1660304062\ldots$. Sampling uses a 7,200-vertex approximation to that smooth domain; its relative area error is 1.50×10^{-7}, far below the experiment's sampling uncertainty. The approximation is disclosed rather than called analytically exact sampling from the smooth curve.

An ellipse would be a weak transfer check, because invertible affine maps preserve all face incidences and multiply both B_n and M_n by the same determinant. The nonelliptical domain was selected to change curvature beyond that trivial invariance.

Domain	n	Graphs	$n^{2/3}\overline{B}_n$	95% graph interval	Prediction
Disk	32	2000	1.16844	[1.14999, 1.18548]	1.23583
Disk	128	2000	1.20784	[1.19413, 1.22233]	1.23583
Disk	512	2000	1.22370	[1.21273, 1.23460]	1.23583
Disk	2048	1500	1.23891	[1.22882, 1.24927]	1.23583
Disk	8192	1000	1.23560	[1.22573, 1.24622]	1.23583
Disk	32768	500	1.23786	[1.22623, 1.24901]	1.23583
Smooth egg	512	1000	1.18769	[1.17182, 1.20334]	1.19688
Smooth egg	2048	1000	1.18408	[1.17054, 1.19638]	1.19688
Smooth egg	8192	750	1.20060	[1.18952, 1.21120]	1.19688
Smooth egg	32768	250	1.18729	[1.17128, 1.20383]	1.19688

Intervals use 2,500 bootstrap replicates of whole graphs. They are not face-level intervals and do not correct simultaneous coverage across all rows. The small-n disk bias is visible and retained. One egg interval at $n = 2048$ narrowly misses the asymptotic reference; no claim that every finite-size interval must cover a limiting constant is made.

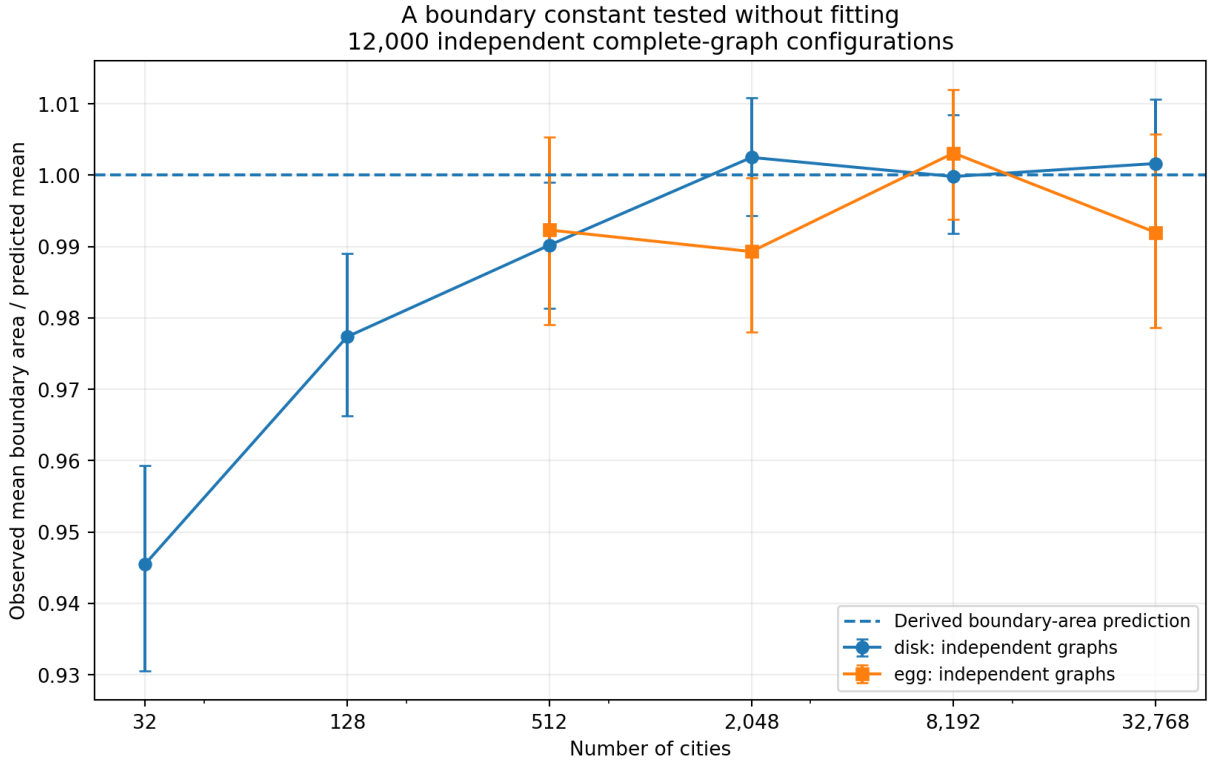


Figure 4: Observed boundary expectation divided by its independently derived prediction. The high-n groups approach one without any fitted correction term.

At $n = 32768$, the ratio of mean boundary area to mean missing area is 0.116322 in the disk, with bootstrap interval $[0.115027, 0.117618]$, versus the prediction 0.116272. For the smooth egg it is 0.115174, with interval $[0.113277, 0.117094]$.

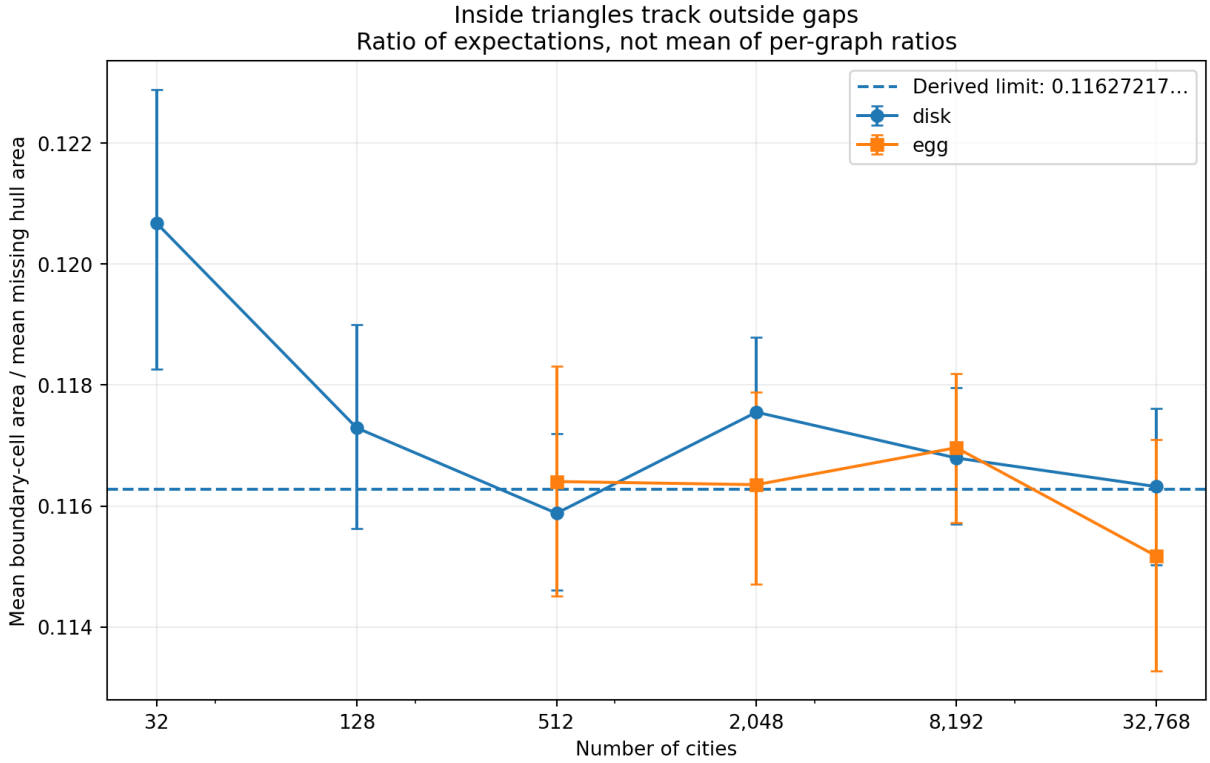


Figure 5: The proposed ratio is between expectations. It is not a deterministic relation between the inside and outside areas of an individual graph.

7.2 A second observable: where the boundary triangles occur

For a fixed disk probe at depth $tn^{-2/3}$, the parabolic calculation also predicts the probability that it belongs to a hull-edge-adjacent face. This is a more detailed test than integrating total area.

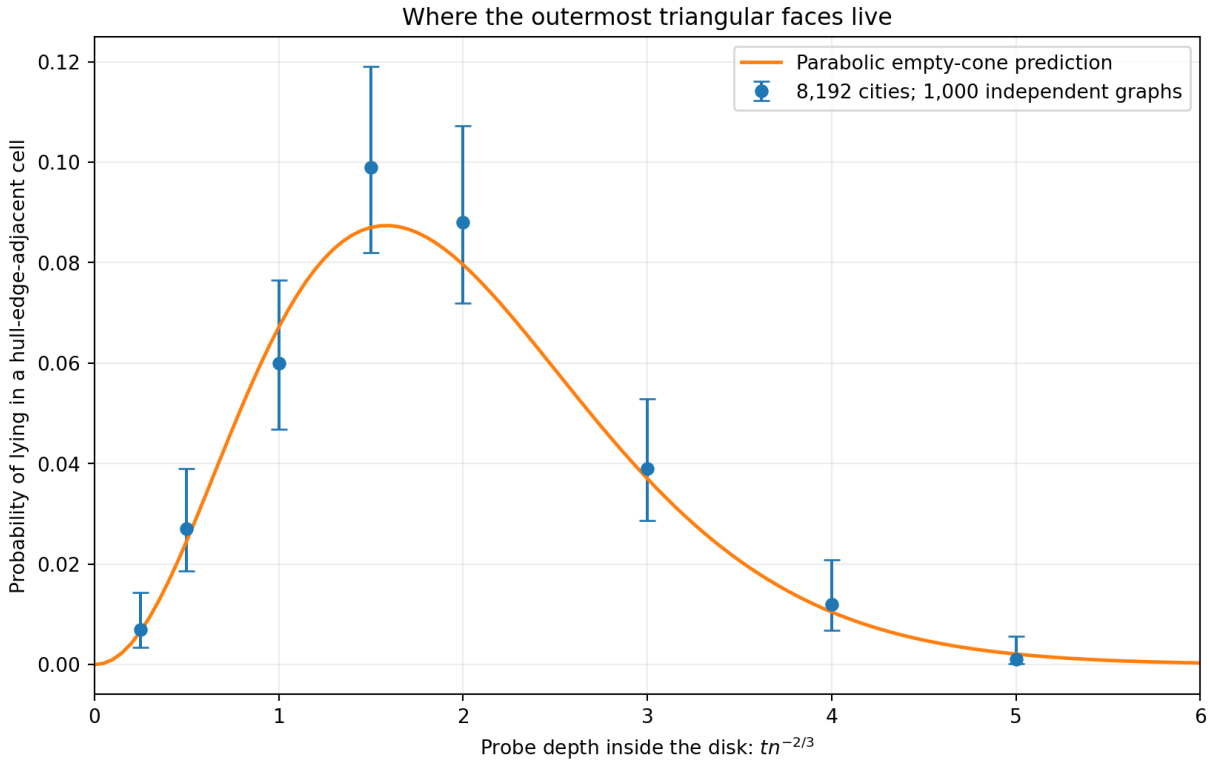


Figure 6: Membership in an outer triangle as a function of inward depth. Error bars are pointwise Wilson binomial intervals over the 1,000 independent 8,192-city graphs. Multiple probes on the same graph are not counted as independent graphs.

For example, at $t = 1.5$ the limiting probability is 0.08702, versus observed 0.099 with interval $[0.08199, 0.11907]$. At $t = 3$ the prediction is 0.03702, versus observed 0.039 with interval $[0.02866, 0.05287]$. The boundary layer contains both exterior hull gaps and interior cells that do not touch a hull edge; these probabilities need not sum to one.

8 How a vanishingly rare class controls large moments

For the disk, the classical hull count satisfies

$$\mathbb{E}h_n \sim 3.3832289658\ldots\, n^{1/3}.$$

Combining the boundary expectation with this yields a pooled boundary-face area scale

$$\frac{\mathbb{E}B_n}{\mathbb{E}h_n} \sim \frac{0.3652798086\ldots}{n}.$$

The quotient is a face-weighted pooled average, not automatically the expected per-graph mean. In the production data, $n\sum B_n/\sum h_n$ is 0.36444 at $n = 8192$ and 0.36735 at $n = 32768$.

The total number of bounded faces grows as order n^4. Thus boundary triangles constitute only order $n^{-11/3}$ of the face count, but their individual area is of order n^{-1}. A small population can therefore affect area-weighted statistics disproportionately.

8.1 A deterministic inequality

Let U be uniform in the sampling domain, independently of the graph, and define $A(U) = 0$ outside its hull. Conditionally on a graph,

$$\mathbb{E}[A(U) \mid P] = \frac{1}{S} \sum_{f \text{ bounded}} A_f^2 \geq \frac{1}{S} \sum_{f \text{ boundary}} A_f^2.$$

Applying Cauchy–Schwarz on the measure consisting of the expected boundary-face counting measure gives

$$\boxed{\mathbb{E}[A(U)] \geq \frac{(\mathbb{E}B_n)^2}{S\mathbb{E}h_n}.} \tag{16}$$

With the derived disk coefficient and the classical hull count, this gives the asymptotic lower-bound coefficient

$$\mathbb{E}[A(U)] \gtrsim 0.1436920864\ldots\, n^{-5/3}.$$

The symbol here records an asymptotic lower bound, not a claim that the full-domain expectation is exactly this constant or even that this is the sharp exponent.

8.2 A lower-order proof independent of the precise constant

The order-$n^{-5/3}$ lower bound can be proved without relying on the numerical value of c_∂. Pack order $n^{1/3}$ disjoint boundary patches of tangential width $O(n^{-1/3})$ and normal depth $O(n^{-2/3})$. In each patch, require one city in each of two small boxes near the parabolic positions $(-1, 1)$ and $(1, 1)$, and no other city in a slightly enlarged finite empty-cone region below a target apex near $(0, 3/2)$. The box and exclusion areas are fixed multiples of $1/n$, so the binomial probability of this event is bounded below by a positive constant for all sufficiently large n.

On the event, the selected pair is a hull edge and its adjacent face has area at least c/n. Different patches produce distinct faces. Summing expectations, with no independence between patch events required, yields

$$\mathbb{E} \sum_{f \text{ boundary}} A_f^2 \geq c' n^{1/3} n^{-2} = c' n^{-5/3}.$$

Consequently

$$\boxed{n^4 \mathbb{E}[A(U)] \geq c'' n^{7/3} \longrightarrow \infty.} \tag{17}$$

This is the important distinction: a fixed-interior local limit does not justify averaging its moments over all locations up to a moving random boundary.

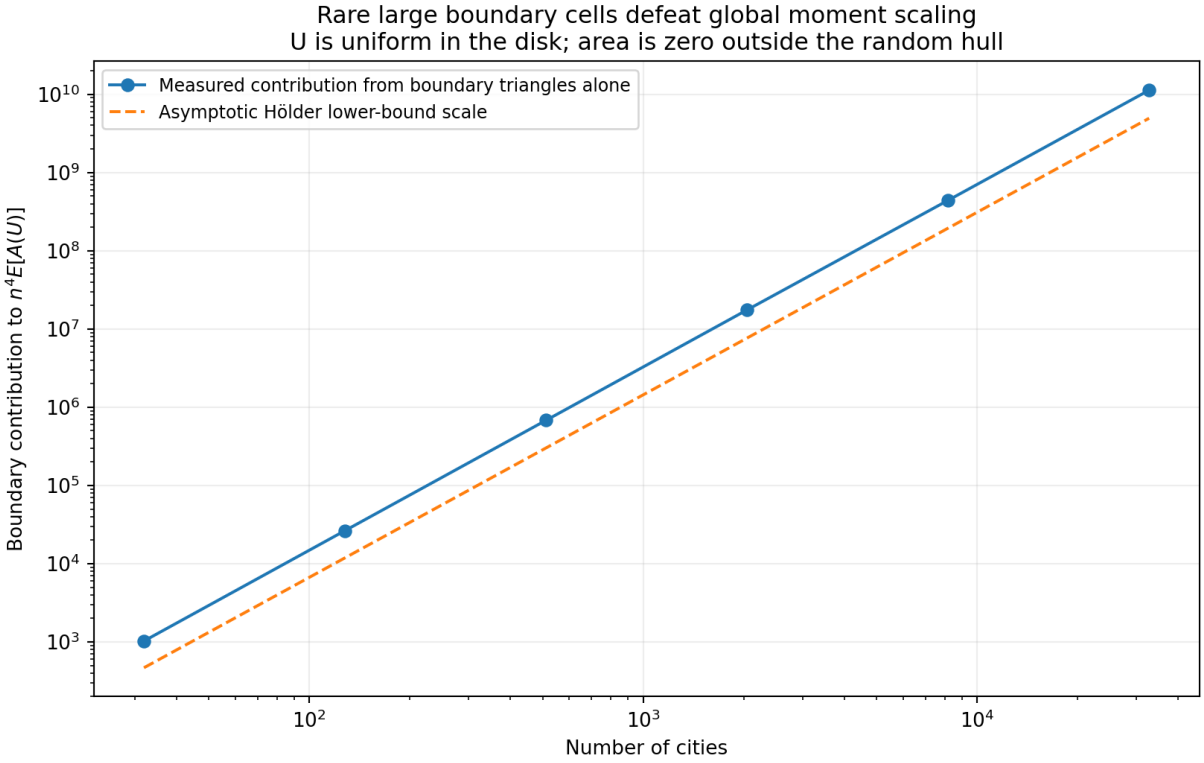


Figure 7: Measured contribution from boundary triangles alone to $n^4\mathbb{E}[A(U)]$. The full-graph quantity is at least this large. The dashed line is a lower-bound scale, not a fitted asymptotic equality.

8.3 A further bound for count-selected faces

For $p \geq 1$, the same patch event gives

$$\mathbb{E} \sum_{f \text{ boundary}} A_f^p \geq c_p n^{1/3-p}.$$

A complete generic graph has at most $\binom{n}{4} + \binom{n}{2} - n + 1 = O(n^4)$ bounded faces. Therefore, for a face selected uniformly from the bounded faces after sampling the graph,

$$\mathbb{E}\left[(n^4 A_{\text{count}})^p\right] \geq c'_p n^{3p-11/3}.$$

In particular these normalized moments must diverge for every $p > 11/9$. This is a sufficient range for boundary-driven divergence, *not* a claim that $11/9$ is a sharp transition: intermediate layers or other rare face types could matter earlier.

8.4 A caution about huge ratios

At $n = 8192$, the measured disk boundary area fraction is approximately 0.0976%. The ratio $\mathbb{E}h_n/\mathbb{E}F_n$ is approximately 5.17×10^{-13}. The all-face denominator uses the exact expected crossing count for uniform disk cities; it is not an enumerated subdivision. These quantities illustrate the disparity, but their ratio is not the mean of a graph-by-graph conditional ratio. The more fundamental statements are the deterministic inequality and the asymptotic powers above.

9 What this changes, and what remains unresolved

The new results explain why one interior random-line model cannot describe every face or every global average. At a fixed interior location, there are many rare pair-generated cuts at scale n^{-2}. At the smooth hull, a few extreme cities remain geometrically decisive, with tangential and normal scales $n^{-1/3}$ and $n^{-2/3}$. The point process in the boundary magnification is Poisson, but shared-endpoint chord dependence does not disappear there as it does in the fixed bulk window.

The exact boundary triangle is determined by two supporting rays, not by the nearest few crossings irrespective of incidence. The new area constant is obtained by counting the endpoint pairs and the regions that must be empty. Its evidence is therefore different from the earlier fractional exponents: it has a specified integral, a curvature transformation, and independent tests at sizes and shapes not used to select its value.

Three limitations are worth preserving. First, the smooth-boundary area asymptotic is given with a localization argument and numerical checks, not an independent referee report. Second, the full intermediate region between hull-scale and fixed-bulk-scale behavior is not characterized here. Third, the moment lower bounds do not identify the exact largest-face law or the sharp asymptotics of every global moment.

Statement	Evidence/status
Every hull edge borders one triangular face	Direct deterministic proof; 772 independent polygonization matches.
Boundary-face probability at a fixed symmetry center equals $n/2^{n-1}$	Direct angular proof; Wendel comparison; 2.4 million angular trials and 480 geometric checks.
Parabolic hull-gap Gamma–Beta representation	Exact limiting-model calculation, classical random-hull context; independent CDF tests.
Boundary/missing-area ratio $0.1162721743\ldots$	Exact parabolic integral; proposed original-graph asymptotic with localization argument; 12,000 independent graphs.
Global n^4-normalized point-selected mean diverges	Deterministic area-bias identity plus a positive-probability cap construction.
Sharp full-area moment or largest-face law	Not established in this note.

10 Reproducibility ledger

The archive's code directory contains the exact supporting-ray routine in `boundary_cells.py`. Other scripts are `validate.py`, `theory.py`, `run_boundary.py`, `exact_center.py`, `summarize.py`, and `make_figures.py`. The seed family for each graph is recorded in the data and is a deterministic function of master seed, domain, city count, and replicate index. No near-coincident-city rejection rule is used.

`data/boundary_graphs_*.csv` contains every production graph record. Compressed face tables are supplied for disk groups up to 8,192 cities. The largest disk and egg groups retain graph-level statistics and deterministic seeds rather than saving every coordinate or face separately. Example point sets are also saved. The theoretical constants and quadrature convergence are in `theoretical_constants.json`; finite validation is in `validation.json`.

The 2.4 million center trials use directions only for the primary classification, because the proof shows that radii are irrelevant to that event. They are distinct from the 12,000 graphs whose complete sets of boundary triangles are actually constructed. The 438,947 triangle count refers only to that production experiment.

No coefficient-fitting code is used in this phase. Bootstrap intervals measure Monte Carlo sampling uncertainty, not all numerical, asymptotic, or model uncertainty. The mean ratio tests use resampled whole graphs. Wilson intervals for fixed boundary probes are pointwise and account for binomial uncertainty; probes at different depths on the same graph are not independent replications.

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